

Approximating the Score Function using Semi-Discrete Optimal Transport

Curves and Surfaces 2026

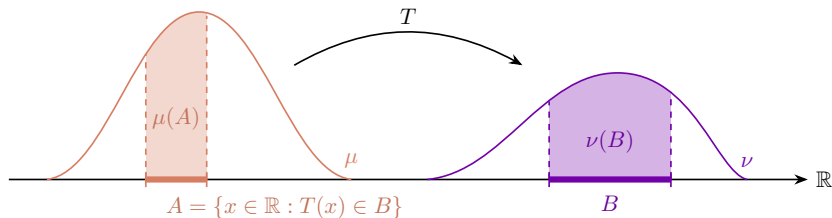
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Laboratoire de Mathématiques d'Orsay

June 11th

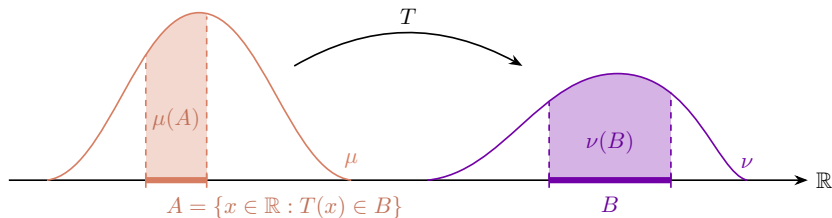
Quick reminders on Optimal Transport 1/2



- ▶ Given ground cost $c(x, y) = \|x - y\|^2$
- ▶ Among all maps T satisfying $\forall B, \nu(B) = \mu(T^{-1}(B))$
- ▶ Find the one minimising the average transport cost

$$W_2(\mu, \nu)^2 := \inf_{T: T_{\#}\mu = \nu} \int_{\mathbb{R}^d} \|x - T(x)\|^2 \mu(dx)$$

Quick reminders on Optimal Transport 2/2

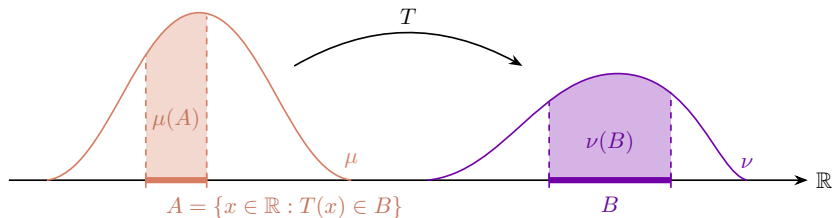


► Kantorovich duality

$$\inf_{T: T_{\#}\mu=\nu} \int_{\mathbb{R}^d} \|x - y\|^2 \mu(dx) = \sup_{\varphi} \int \varphi(x) \mu(dx) + \int \varphi^c(y) \nu(dy),$$

where $\varphi^c(y) := \inf_x c(x, y) - \varphi(x)$.

Quick reminders on Optimal Transport 2/2



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► Let $c(x, y) = \alpha \|x - y\|^2$

$$\nabla_x c(x, T^*(x)) = \nabla \varphi^*(x) \iff T^*(x) - x = \boxed{y - x = -\frac{1}{2\alpha} \nabla \varphi^*(x)}$$

Displacement

What is the score function?

- **Forward process:** Ornstein–Uhlenbeck (OU) process:

$$dX_t = -X_t dt + \sqrt{2} d\omega_t, \quad X_0 \sim \mu, \quad t \in [0, T].$$

Let $\rho_t := \text{Law}(X_t)$. Then ρ_t solves the Fokker–Planck equation

$$\partial_t \rho_t(x) = \Delta \rho_t(x) + \text{div}(\rho_t(x)x). \quad (\star)$$

whose stationary solution is $\gamma \propto e^{-\|\cdot\|^2/2}$.

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- **Macroscopic POV:** Equation $(\star)$ is the 2-Wasserstein Gradient Flow of the relative entropy $\mathcal{H}(\rho|\gamma) := \int \frac{\|x\|^2}{2} \rho + \int \rho \ln(\rho)$. It means $(\star)$ can be written as

$$\partial_t \rho_t(x) = -\nabla \cdot (\rho_t(x)v_t(x)),$$

where $v_t(x) = -x - \nabla \ln(\rho_t(x))$.

Using OT to learn the score 1/2

- ▶ Given a time-step $\tau > 0$, such a gradient flow can be discretised in time using the **JKO scheme** (or Implicit Euler scheme, or Proximal step)

$$\begin{cases} \rho_0 = \mu \\ \rho_{n+1}^\tau := \arg \min \mathcal{H}(\rho|\gamma) + \frac{1}{2\tau} W_2(\rho_n^\tau, \rho)^2 \end{cases}$$

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- ▶ **At step n :** We show that

$$\rho_{n+1}^\tau \propto \exp\left(-(\varphi^\star)^{c/2\tau}\right) d\gamma$$

where $\varphi^\star$ is a Kantorovich potential satisfying

$$\varphi^\star \in \arg \min_{\varphi} \tilde{J}(\varphi) := \ln \left(\int e^{-\varphi^{c/2\tau}} d\gamma \right) - \int \varphi d\rho_n^\tau.$$

Using OT to learn the score 2/2

- ▶ The optimal potential $\varphi^\star$ satisfies ($\alpha = \frac{1}{2\tau}$ here)

$$-\tau \nabla \varphi^\star(x) = y - x$$

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- ▶ The displacement of the particle $\boldsymbol{x}$ being $\tau v_t(\boldsymbol{x})$, we obtain

$$-\tau \nabla \varphi^\star(\boldsymbol{x}) = \tau v_t(\boldsymbol{x}) = -\tau(\boldsymbol{x} + \nabla \ln(\rho_{n+1}^\tau(\boldsymbol{x}))) \iff \boxed{\nabla \ln(\rho_{n+1}^\tau(\boldsymbol{x})) = \nabla \varphi^\star(\boldsymbol{x}) - \boldsymbol{x}}.$$

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- ▶ We'll approximate the score by approximating the Kantorovich potential $\varphi^\star$, that minimises the functional:

$$\tilde{J}(\varphi) = \ln \left(\int e^{-\varphi^c/2\tau} \right) d\boldsymbol{\gamma} - \int \varphi d\rho_n^\tau$$

General contributions

Proposition (G.)

The functional $\tilde{J}(\varphi) := \ln \left(\int e^{-\varphi^{c/2\tau}} d\gamma \right) - \int \varphi d\rho_n^\tau$ is convex.

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Theorem (G.)

Let $J(\varphi) := \int e^{-\varphi^{c/2\tau}} d\gamma - \int \varphi d\rho_n^\tau$. Then,

$$\arg \min_{\varphi} J(\varphi) \subseteq \arg \min_{\varphi} \tilde{J}(\varphi) \quad \text{and} \quad \arg \min_{\int e^{-\varphi^{c/2\tau}} d\gamma = 1} \tilde{J}(\varphi) = \arg \min_{\varphi} J(\varphi).$$

Pro: J is "SGDisable"

Con: one more parameter to estimate
(the normalisation constant!)

Stochastic gradient descent 1/3

$$\blacktriangleright J(\varphi) = \mathbb{E}_{\mathbf{y} \sim \gamma} \left[\exp \left(-\varphi^{c/2\tau}(\mathbf{y}) \right) - \int \varphi(\mathbf{x}) \rho_n^\tau(\mathrm{d}\mathbf{x}) \right]$$

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- ▶ Choice of parametrisation: semi-discrete optimal transport using

$$\hat{\rho}_n^\tau = \frac{1}{N} \sum_{i=1}^N \delta_{\hat{x}_i^{(n)}} \rightsquigarrow \begin{cases} \varphi(x; \theta) = \theta^\top \varphi(x) \\ \varphi(x) = (\varphi_1(x), \dots, \varphi_N(x)) \\ \varphi_i(x) = \mathbb{1}_{\{x = \hat{x}_i^{(n)}\}} \end{cases}$$

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► $G(\theta) := J(\varphi(\cdot; \theta)) = \mathbb{E}_{\mathbf{y} \sim \gamma} \left[\exp \left(- \min_{1 \leq i \leq N} \left\{ \frac{c(\hat{\mathbf{x}}_i^{(n)}, \mathbf{y})}{2\tau} - \theta_i \right\} \right) - \frac{1}{N} \sum_{i=1}^N \theta_i \right]$

Stochastic gradient descent 2/3

Projected SGD algorithm ($c_\tau(\mathbf{x}, \mathbf{y}) := \frac{c(\mathbf{x}, \mathbf{y})}{2\tau}$) for some R_{proj} :

$$\left\{ \begin{array}{l} \text{pick } \mathbf{y} \sim \gamma \\ \text{let } i := \arg \min_j c_\tau(\hat{\mathbf{x}}_j^{(n)}, \mathbf{y}) - \theta_j \\ \text{define } e_i \text{ as the } i\text{-th canonical basis vector of } \mathbb{R}^N \\ \theta^{(k+1)} := \text{proj}_{B(0, R_{\text{proj}})} \left(\theta^{(k)} - \alpha_k \underbrace{e^{\theta_i^{(k)} - c_\tau(\hat{\mathbf{x}}_j^{(n)}, \mathbf{y})}}_{\text{A priori unbounded!}} e_i + \frac{\alpha_k}{N} \mathbf{1} \right) \end{array} \right.$$

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► Gradient bounded by $e^{R_{\text{proj}}}$

Stochastic gradient descent 3/3

Theorem (G .)

► $\alpha_k = \frac{C}{\sqrt{k}}, \quad \hat{\theta}_K := \frac{1}{K} \sum_{k=0}^K \theta^{(k)}$

Then, it holds that

$$0 \leq \mathbb{E}[G(\hat{\theta}_K)] - G(\theta^*) \leq \frac{2R_{\text{proj}}^2}{CK} + \frac{B^2C}{\sqrt{K}} = \mathcal{O}(K^{-\frac{1}{2}}),$$

where θ^* denotes the unique minimiser of G .

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Approximating the next JKO iterate

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Theoretical JKO $_{\tau}$ $\rightarrow$ $\tilde{\rho}_{n+1}^{\tau}(\mathrm{d}y) \propto \max_i e^{\theta_i - c_{\tau}(\hat{x}_i^{(n)}, y)} \gamma(\mathrm{d}y)$

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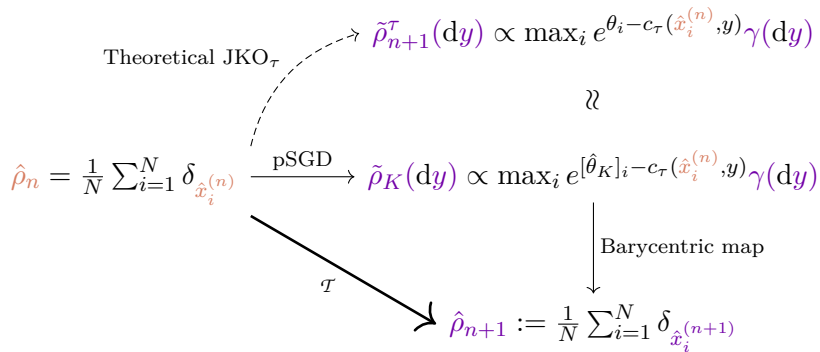
Approximating the next JKO iterate

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Approximating the next JKO iterate

$$\begin{array}{ccc}
 & \text{Theoretical JKO}_\tau & \nearrow \tilde{\rho}_{n+1}^\tau(dy) \propto \max_i e^{\theta_i - c_\tau(\hat{x}_i^{(n)}, y)} \gamma(dy) \\
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 & & \downarrow \text{Barycentric map} \\
 & & \hat{\rho}_{n+1} := \frac{1}{N} \sum_{i=1}^N \delta_{\hat{x}_i^{(n+1)}}
 \end{array}$$

Approximating the next JKO iterate



More on the barycentric map

- ▶ $\tilde{\rho}_K(\mathrm{d}y) \propto \max_i e^{[\hat{\theta}_K]_i - c_\tau(\hat{x}_i^{(n)}, y)} \gamma(\mathrm{d}y)$ is **piecewise log-quadratic**

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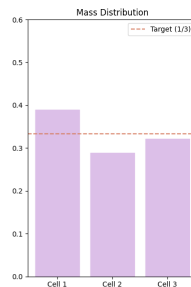
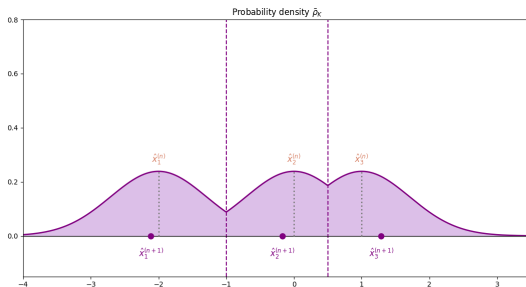
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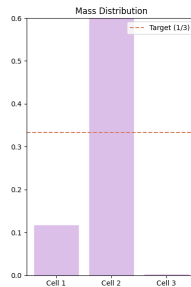
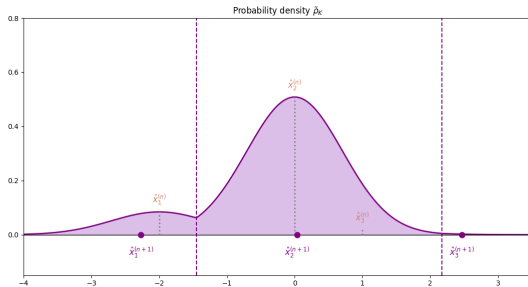
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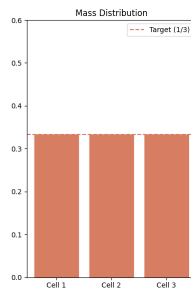
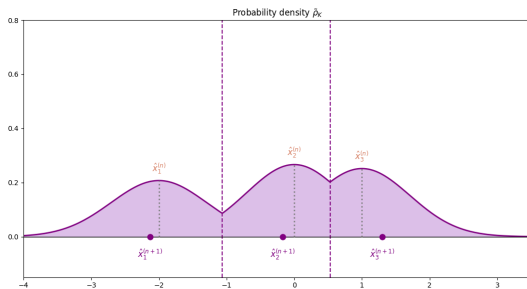


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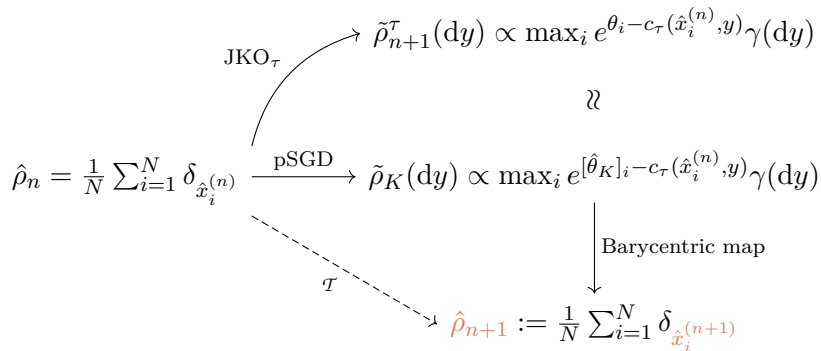
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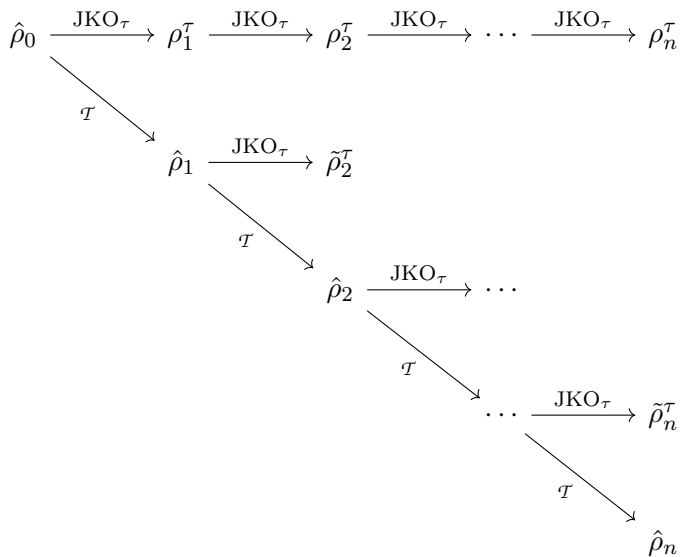
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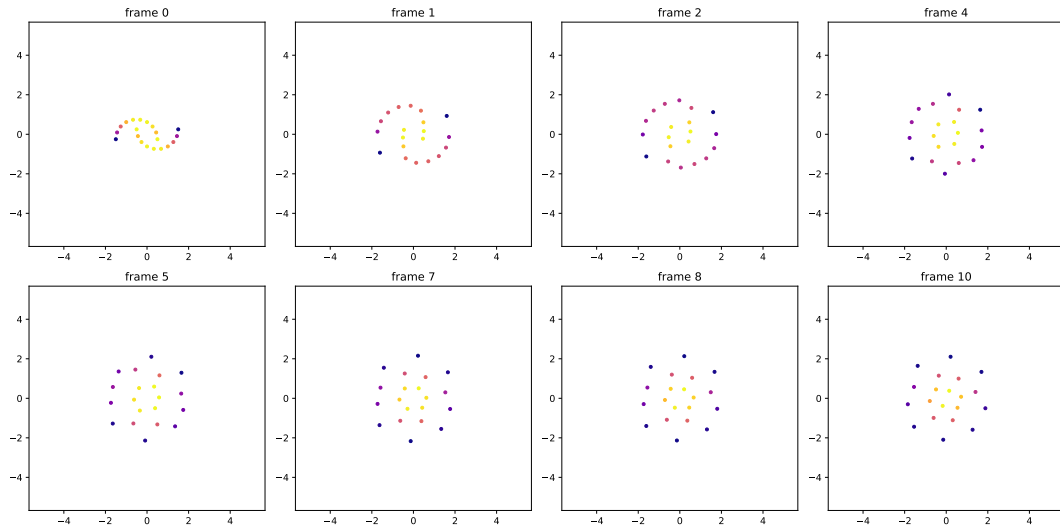
Approximating the next JKO iterate



Iterating the whole procedure



Some “animations”



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- ▶ The true score is piecewise linear

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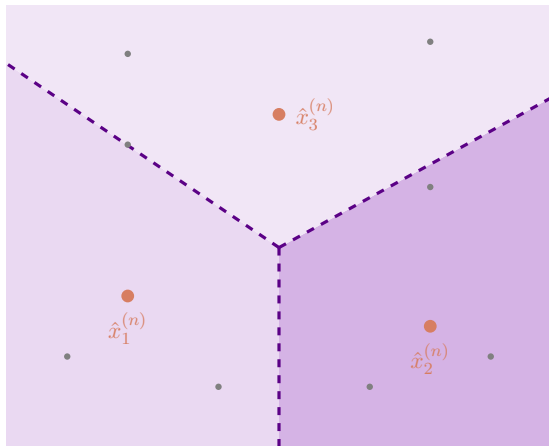
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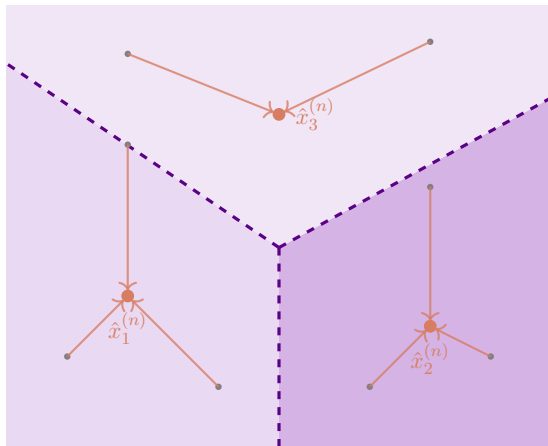
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Thank you for your attention!